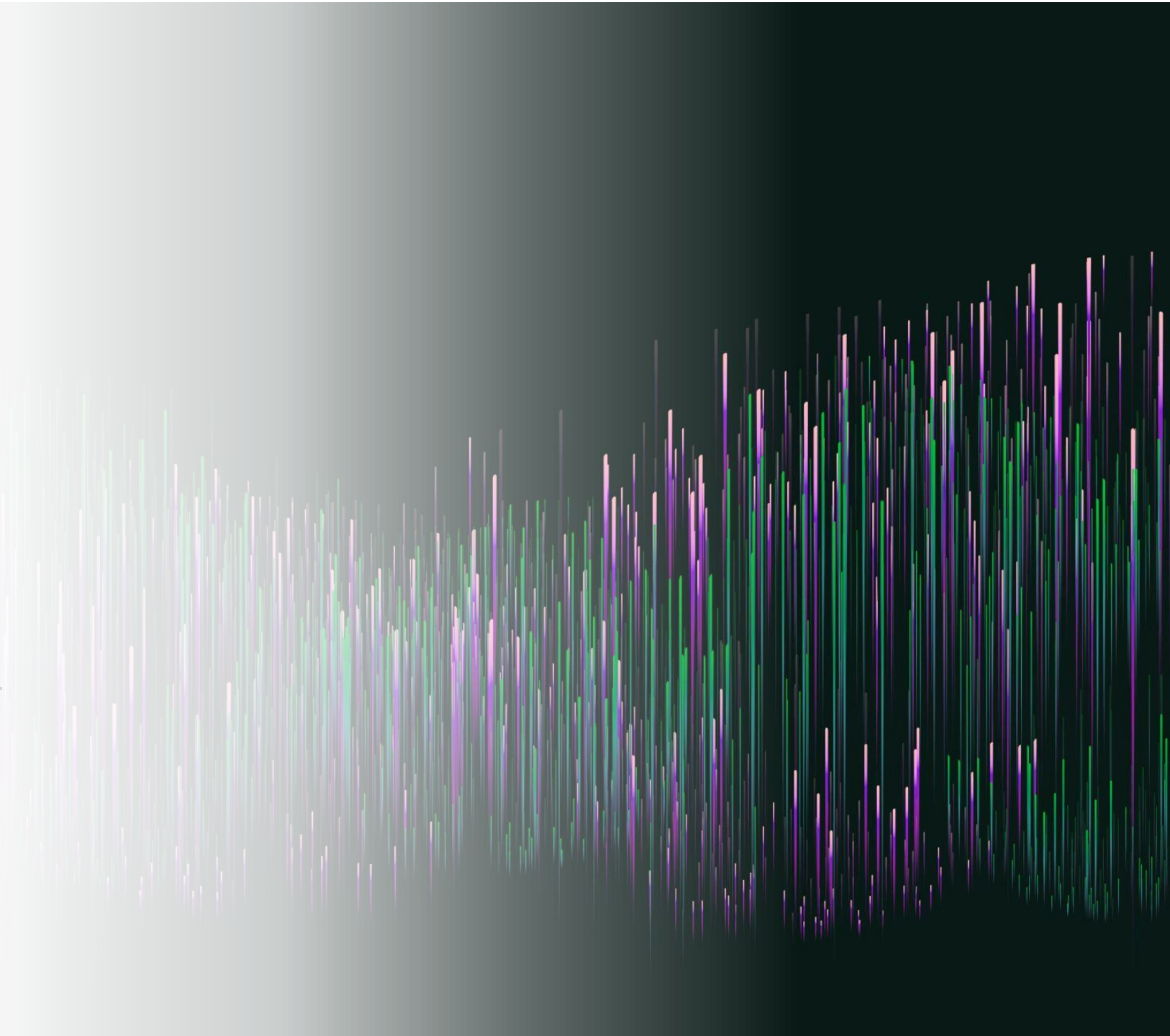




Network Topologies

By

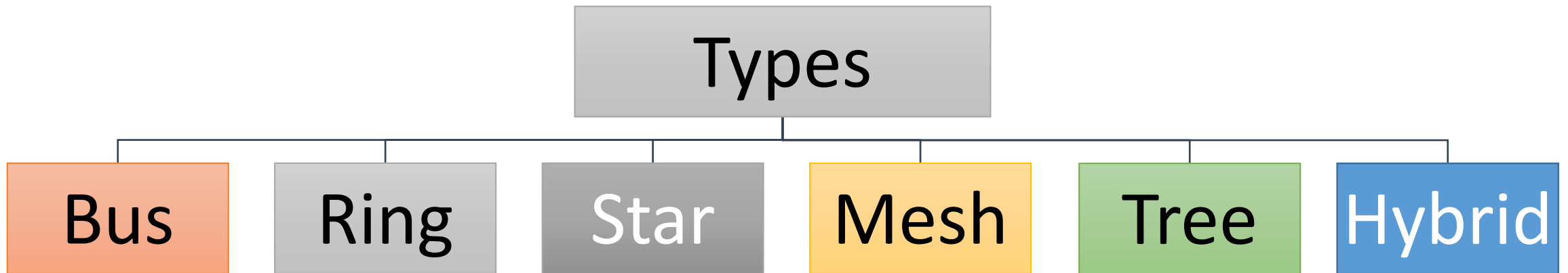
Dr Sanjeev Kumar Mandal



What is topology?

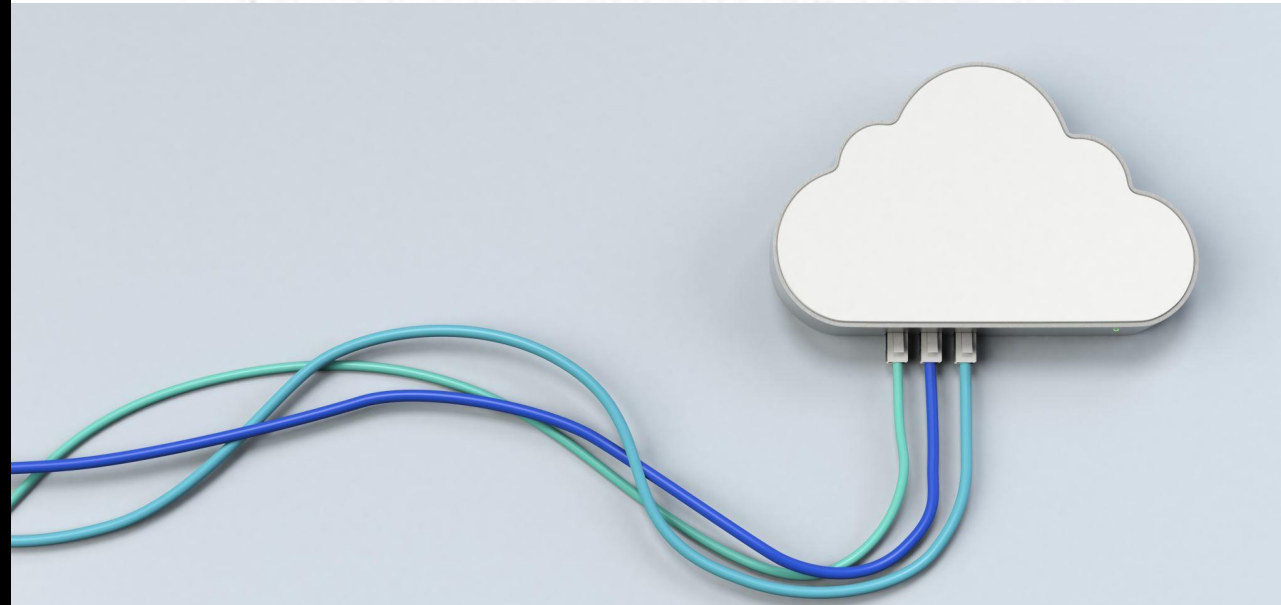
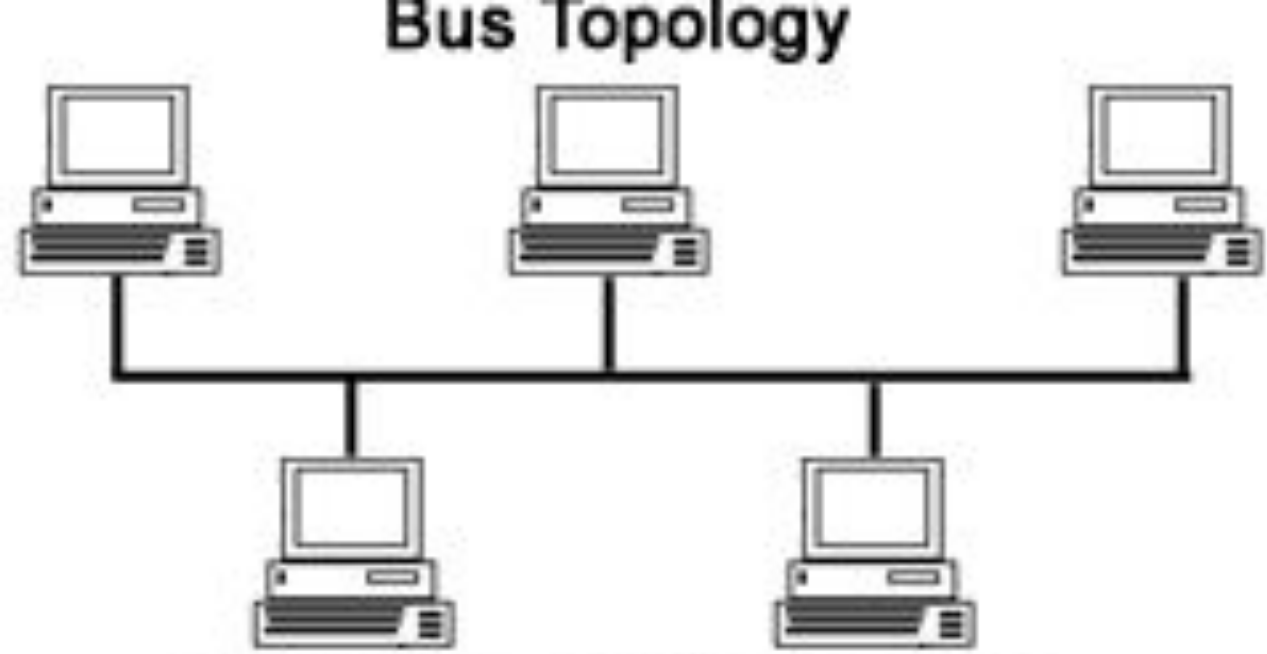
- A physical arrangement of computer systems or nodes that are connected through communication media is called topology.
- There are six different types of topology:

Topologies

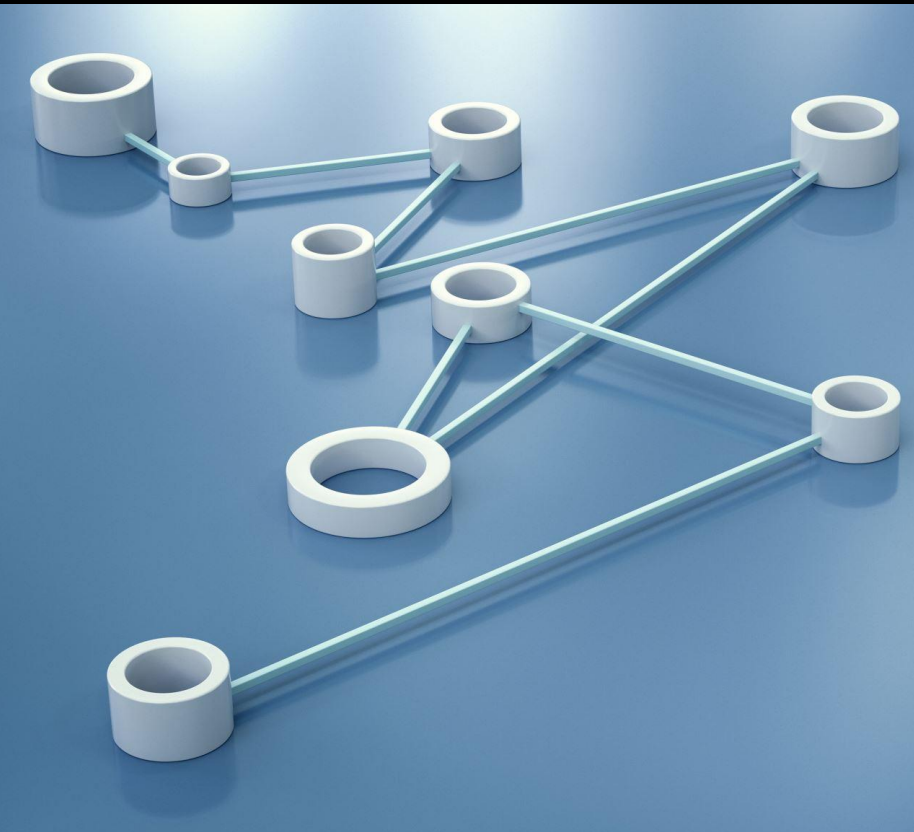


Bus Topology

- In bus topology one long cable act as a single communication and all the devices are connected to this cable.
- Alternatively referred to as **line topology**
- In Bus topology all the computer are connected parallel.
- Every Node have a subnet of the server.

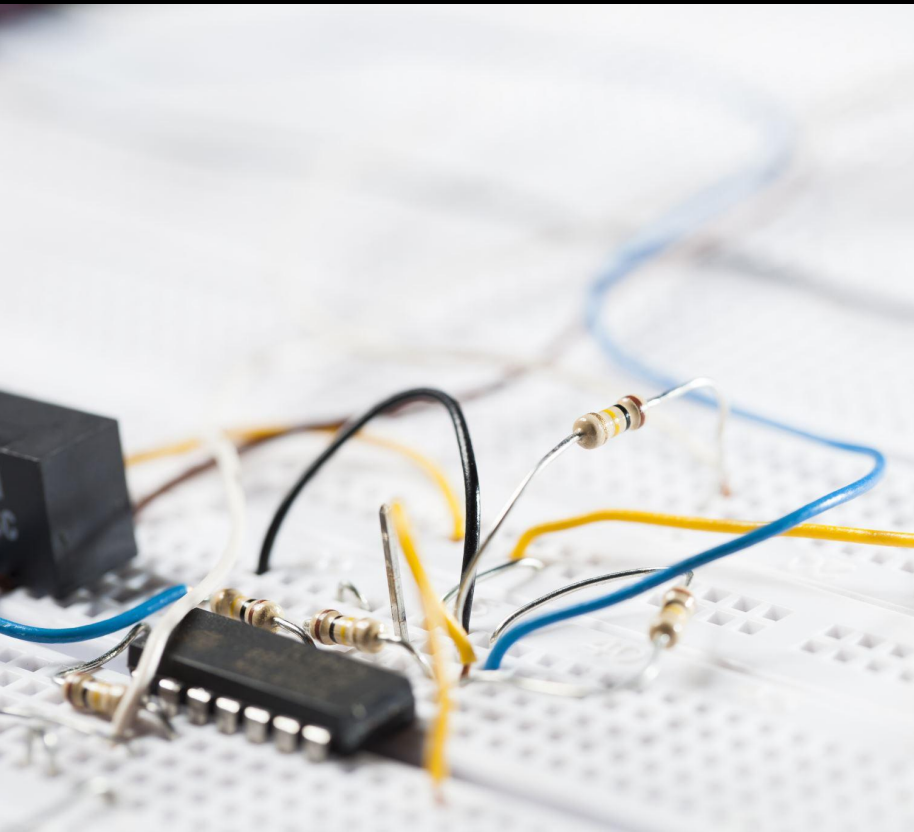


Advantages



1. It works well when you have a small network.
2. It's the easiest network topology for connecting computers or peripherals in a linear fashion.
3. It requires less cable length than a star topology.
4. Less Expensive
5. If one computer fails to work it will not affect other computers.

Disadvantages of bus topology



- 1. It can be difficult to identify the problems if the whole network goes down.**
- 2. It can be hard to troubleshoot individual device issues.**
- 3. Bus topology is not great for large networks.**
- 4. Terminators are required for both ends of the main cable.**
- 5. Additional devices slow the network down.**
- 6. If a main cable is damaged, the network fails or splits into two.**

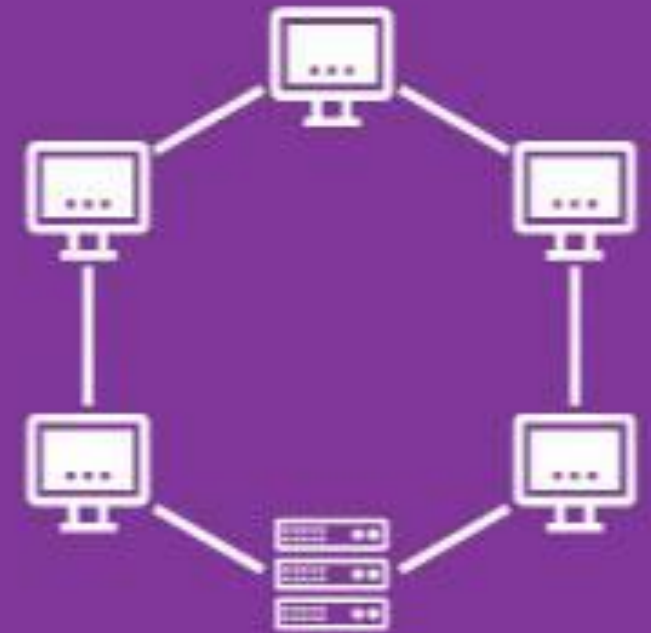
Ring Topology

Ring topology is a type of network topology in which each device is connected to two other devices on either side via an RJ-45 cable or coaxial cable.

This forms a circular ring of connected devices which gives it its name.

Data is commonly transferred in one direction along the ring, known as a unidirectional ring.

The data is forwarded from one device to the next, until it reaches the intended destination. In a bidirectional ring, data can travel in either direction.



Types of Ring topology

- There are multiple types of ring topologies that provide redundancy, including:
- **Counter-rotating ring topology.** An additional ring is present in the opposite direction of the initial ring. If a network switch fails, the backup ring is activated which allows data to continue to transmit in the network.
- **Collapsed ring topology.** If a device fails, the ring will “collapse” and connect every other device to each other, eliminating the single point of failure and isolating it for easier maintenance.

Advantages

- Since data flows in one direction, the chance of a packet collision is reduced
- A network server is not needed to control network connectivity
- Devices can be added without impacting network performance
- Easy to identify and isolate single points of failure
- Better suited for high traffic environments than a [bus topology](#)

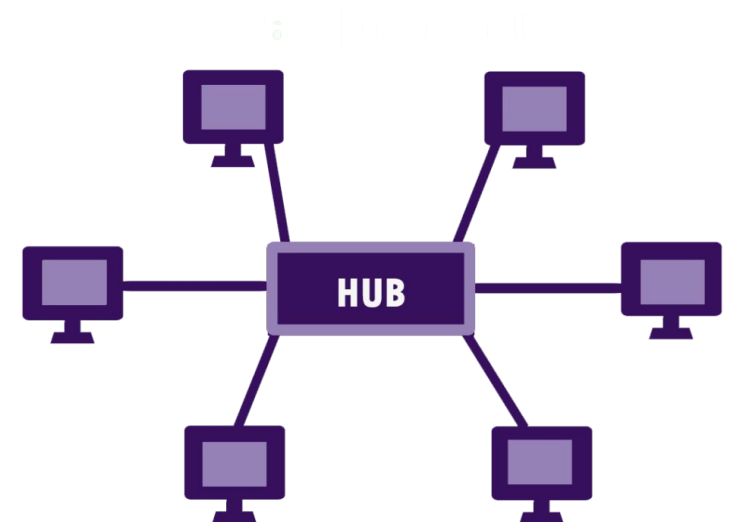
Disadvantages



- All data travelling over the network must pass through each device on its way to its destination, which can reduce performance
- If one device fails, the entire network is impacted
- Can be difficult to architect the necessary cabling
- More expensive to implement than a bus topology

Star topology

- Alternatively referred to as a **star network**, **star topology** is one of the most common [network](#) setups. Every [node](#) connects to a central network device in this configuration, like a [hub](#), [switch](#), or computer.
- The central network device acts as a [server](#), and the peripheral devices act as [clients](#).
- In a star topology setup, either a [coaxial](#) or [RJ-45](#) network cable is used, depending on each computer's type of [network card](#).
- The image shows how this network setup gets its name, as it is shaped like a star.



Advantages

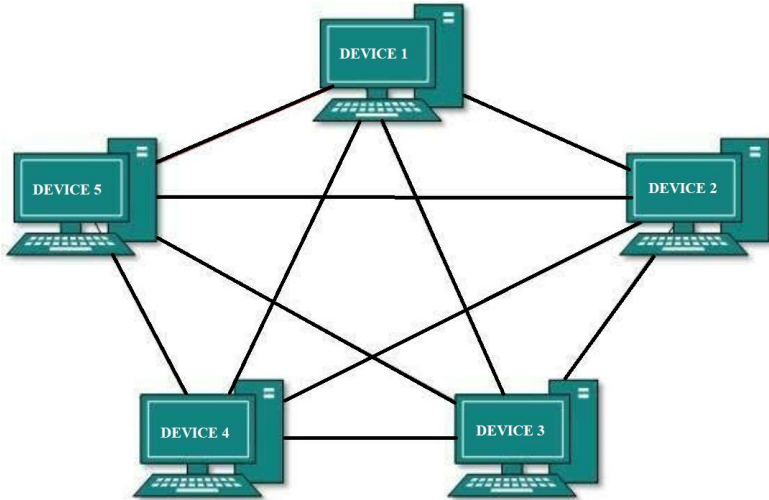
- Centralized management of the network through the use of the central computer, hub, or switch.
- Easy to add another computer to the network.
- If one computer on the network fails, the rest of the network continues to function normally.



Disadvantages

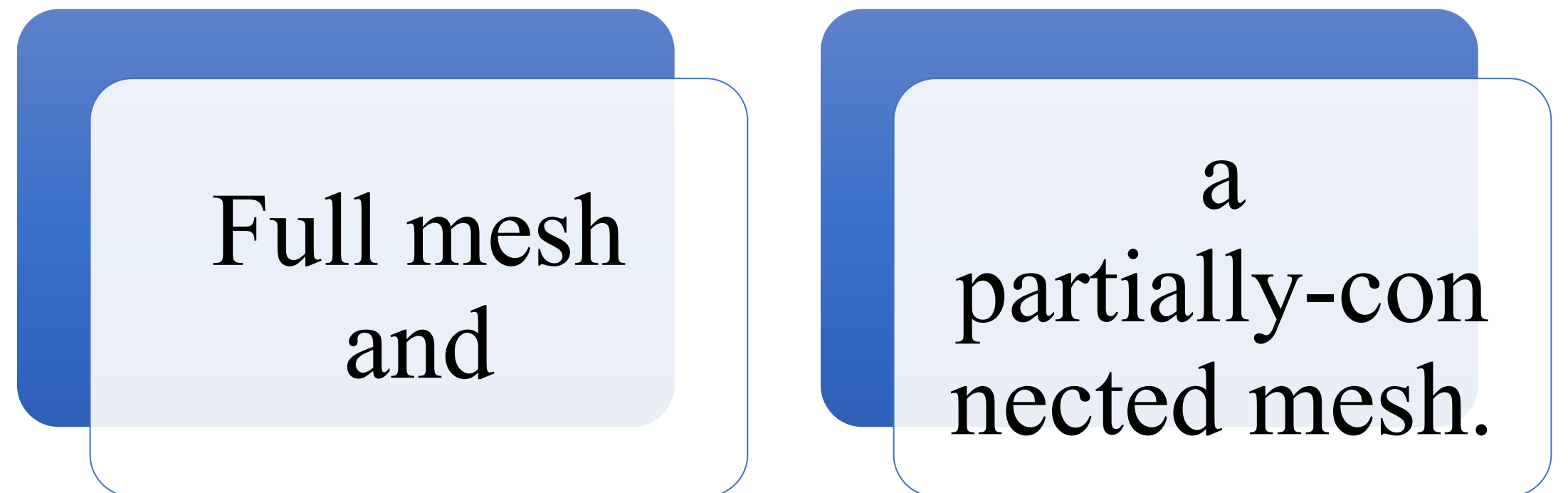
- It may have a higher cost to implement, especially when using a switch or router as the central network device.
- The central network device determines the performance and number of nodes the network can handle.
- If the central computer, hub, or switch fails, the entire network goes down, and all computers are disconnected from the network.

Mesh topology



- A mesh topology is a network setup where each computer and network device is interconnected with one another.
- This topology setup allows for most transmissions to be distributed even if one of the connections goes down. It is a topology commonly used for [wireless networks](#).
- Below is a visual example of a simple computer setup on a network using a **mesh topology**.

Types of mesh topology



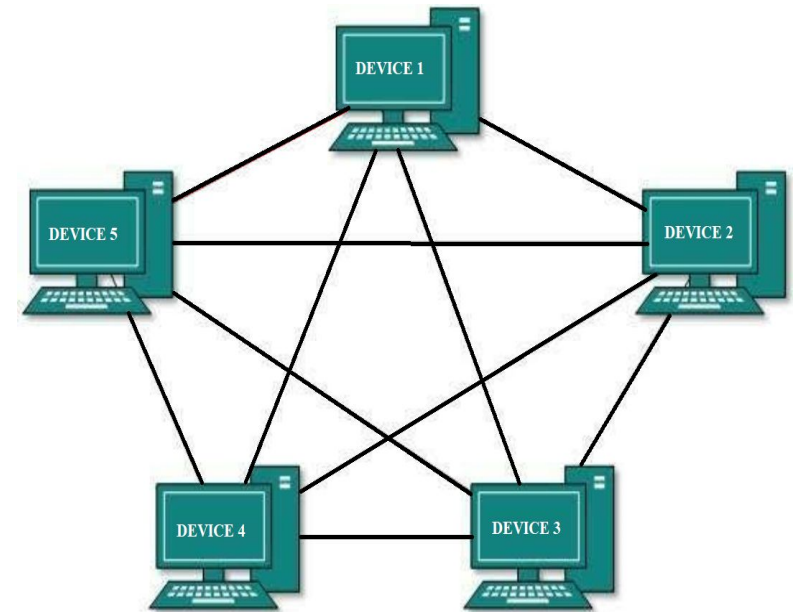
The diagram consists of two identical graphic elements side-by-side. Each element features a dark blue rounded rectangle with a light blue rounded rectangle centered inside it. The text is placed within the light blue area.

Full mesh
and

a
partially-con
nected mesh.

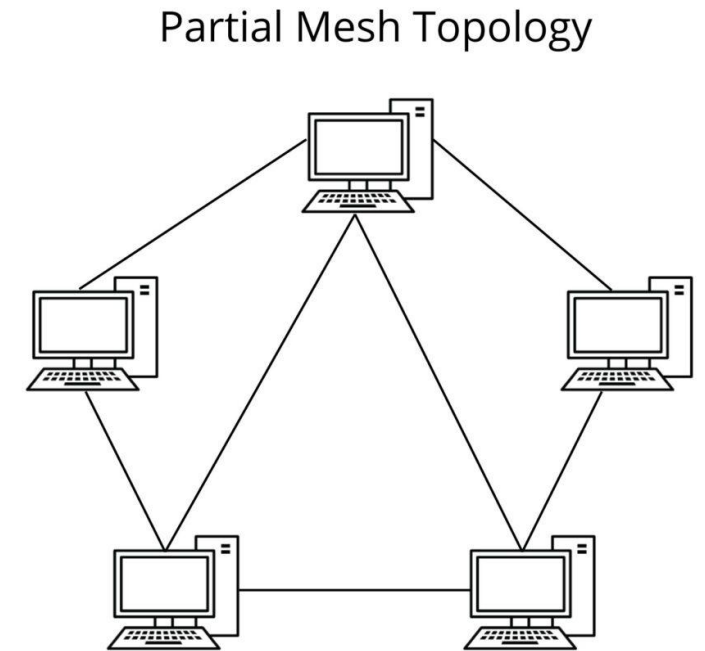
Full mesh

- In a *full mesh topology*, every computer in the network has a connection to each of the other computers in that network.
- The number of connections in this network can be calculated using the following formula (n is the number of computers in the network): $n(n-1)/2$



Partially mesh

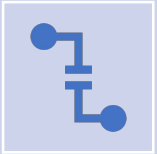
- In a ***partially-connected mesh topology***, at least two of the computers in the network have connections to multiple other computers in that network.
- It is an inexpensive way to implement redundancy in a network. If one of the primary computers or connections in the network fails, the rest of the network continues to operate normally.



Advantages



Manages high amounts of traffic, because multiple devices can transmit data simultaneously.



A failure of one device does not cause a break in the network or transmission of data.

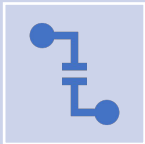


Adding additional devices does not disrupt data transmission between other devices.

Disadvantages



The cost to implement is higher than other network topologies, making it a less desirable option.



Building and maintaining the topology is difficult and time consuming.

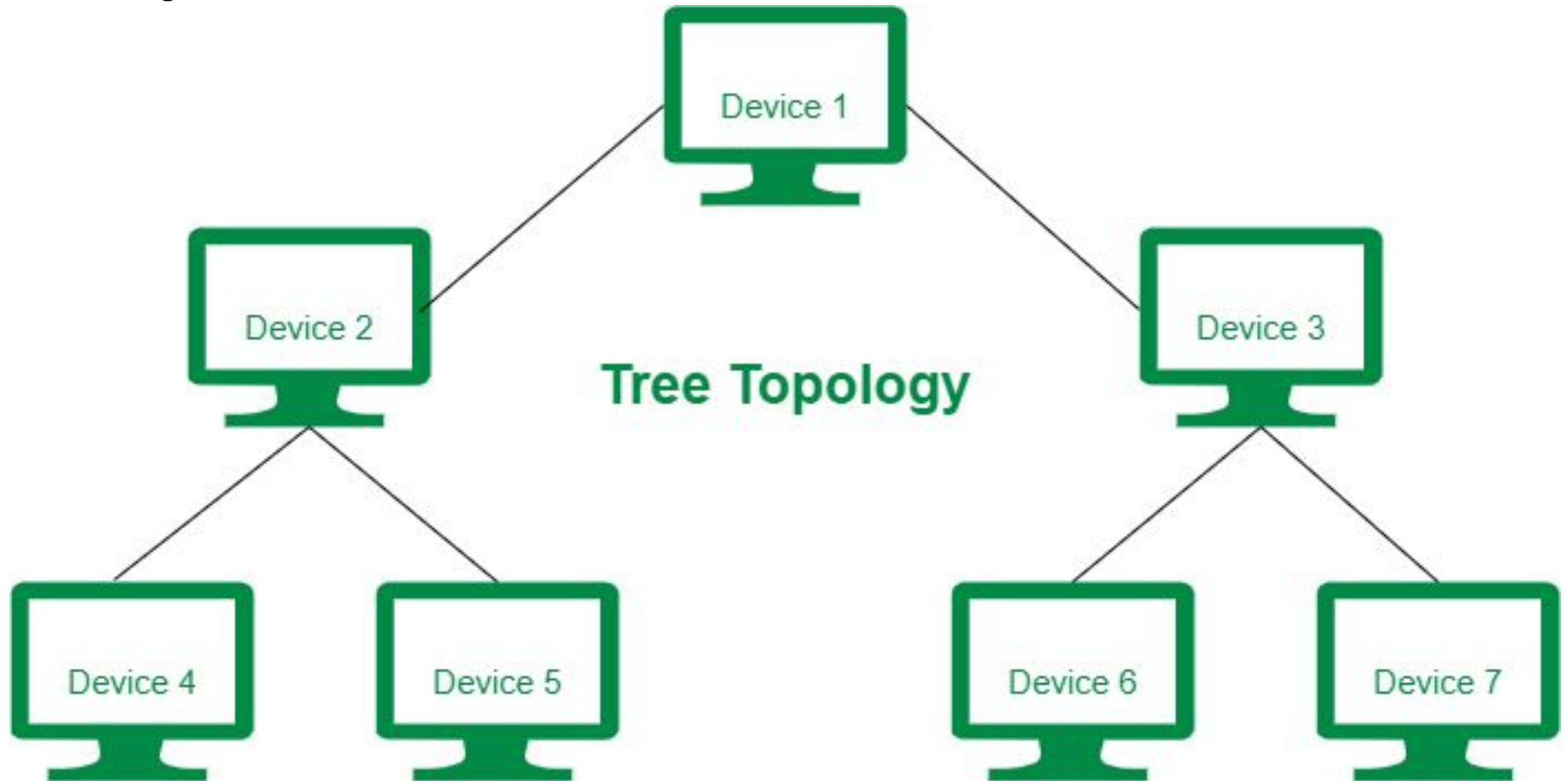


The chance of redundant connections is high, which adds to the high costs and potential for reduced efficiency.

Tree topology

- A **tree topology** is a special type of structure where many connected elements are arranged like the branches of a tree.
- For example, tree topologies are frequently used to organize the computers in a corporate [network](#), or the information in a [database](#).
- In a tree topology, there can be only one connection between any two connected nodes.
- Because any two nodes can have only one mutual connection, tree topologies create a natural [parent and child](#) hierarchy.

Example



Hybrid topology

- A **hybrid topology** is a type of network topology that uses two or more differing network [topologies](#).
- These topologies can include a mix of [bus topology](#), [mesh topology](#), [ring topology](#), [star topology](#), and [tree topology](#).
- The choice to use a hybrid topology over a standard topology depends on the needs of a business, school, or the users.
- The number of computers, their location, and desired network performance are all factors in the decision.

Example

